

Challenge (Habanero)

- Q1.** What type of plot is used to display the distribution of 24, 15, 36, 27, 18?
(A) Box plot
(B) Stem-and-leaf plot
(C) Histogram
(D) Scatter plot
(E) Bar graph
- Q2.** The times (in minutes) for 5 hikers to complete a trail are 23, 17, 32, 19, 28. What is the stem for 23 in a stem-and-leaf plot?
(A) 2
(B) 3
(C) 20
(D) 30
(E) 10
- Q3.** A stem-and-leaf plot shows fuel consumption (in miles per gallon) for 6 vehicles. If a vehicle has a fuel consumption of 27 miles per gallon, what is the leaf?
(A) 20
(B) 7
(C) 2
(D) 27
(E) 30
- Q4.** Which of the following distributions is symmetrical about the median?
(A) Skewed left
(B) Skewed right
(C) Uniform
(D) Bimodal
(E) Normal
- Q5.** What is the purpose of a stem-and-leaf plot?
(A) To compare two datasets
(B) To display the distribution of a dataset
(C) To calculate the mean and median
(D) To visualize the correlation between two variables
(E) To create a box plot
- Q6.** If a stem-and-leaf plot has the stem 2 with leaves 1, 3, 5, 7, 9, what is the range of the data?
(A) 8
(B) 10
(C) 12
(D) 16
(E) 20
- Q7.** A wildlife researcher records the weights (in pounds) of 5 deer: 21, 18, 25, 22, 19. What would the stem-and-leaf plot look like?
(A) 1|8, 9 2|1, 2, 5
(B) 1|8, 9 2|1, 2, 5, 8
(C) 2|1, 2, 5, 8, 9
(D) 1|8 2|1, 2, 5, 8, 9
(E) 1|9 2|1, 2, 5, 8, 9
- Q8.** The number of birds in 6 different regions is 15, 28, 32, 18, 25, 20. If these values are plotted on a stem-and-leaf plot, which region has the most birds?
(A) Region with stem 1 and leaf 5
(B) Region with stem 2 and leaf 8
(C) Region with stem 3 and leaf 2
(D) Region with stem 2 and leaf 5
(E) Region with stem 3 and leaf 5

Open Ended Questions (FRQ)

- Q9.** Create a stem-and-leaf plot for the following distances (in miles) traveled by 8 hikers: 12, 18, 24, 20, 15, 22, 19, 25. Interpret the distribution and describe any patterns or trends in the data.
- Q10.** Suppose a park ranger collects data on the ages (in years) of 10 animals in a wildlife sanctuary: 3, 5, 2, 8, 1, 9, 4, 7, 6, 3. Use a stem-and-leaf plot to display the distribution of the ages and calculate the range of the data. Show your work and explain your answer.

Chef's Tips

- Tip 1: Ensure students understand how to construct and interpret stem-and-leaf plots.
- Tip 2: Provide clear instructions and examples for students to work on.
- Tip 3: Encourage students to analyze and describe patterns or trends in the data.